

① 單元 1 坐標系與函數圖形

(主題 1~13: 實數與絕對值、直角坐標系)

答案

- | | |
|-----------------------|---------------------|
| 1. $30+2\sqrt{2}$ | 2. $50+550\sqrt{3}$ |
| 3. 48 | 4. 3 或 -8 |
| 5. $-3 \leq x \leq 2$ | 6. $3\sqrt{2}$ |
| 7. 3 | 8. 7 |
| 9. (0,2) | 10. (3,3) |

$$1. \because \sqrt{81} = \sqrt{9^2} = 9, \sqrt{64} = \sqrt{8^2} = 8$$

$$\sqrt{169} = \sqrt{13^2} = 13$$

$$\therefore a = 9 + 8 + 13 = 30$$

$$\because \sqrt{50} = \sqrt{5^2 \times 2} = 5\sqrt{2}$$

$$\sqrt{98} = \sqrt{7^2 \times 2} = 7\sqrt{2}$$

$$\sqrt{200} = \sqrt{10^2 \times 2} = 10\sqrt{2}$$

$$\therefore b = 5\sqrt{2} + 7\sqrt{2} - 10\sqrt{2} = 2\sqrt{2}$$

$$\text{故 } a+b = 30 + 2\sqrt{2}$$

$$2. a = \frac{300 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{300\sqrt{3}}{3} = 100\sqrt{3}$$

$$b = \frac{400 \times (\sqrt{3}-1)}{(\sqrt{3}+1) \times (\sqrt{3}-1)} = \frac{400(\sqrt{3}-1)}{2}$$

$$= 200(\sqrt{3}-1)$$

$$c = \frac{500 \times (\sqrt{3}+1)}{(\sqrt{3}-1) \times (\sqrt{3}+1)} = \frac{500(\sqrt{3}+1)}{2}$$

$$= 250(\sqrt{3}+1)$$

$$\text{故 } a+b+c = 100\sqrt{3} + 200(\sqrt{3}-1) + 250(\sqrt{3}+1)$$

$$= 100\sqrt{3} + 200\sqrt{3} - 200 + 250\sqrt{3} + 250$$

$$= 50 + 550\sqrt{3}$$

$$3. \text{ 算幾不等式: } \frac{3a+b}{2} \geq \sqrt{3a \times b}$$

$$\Rightarrow \frac{24}{2} \geq \sqrt{3ab} \xrightarrow{\text{平方}} 144 \geq 3ab$$

$$\Rightarrow ab \leq 48$$

因此 ab 的最大值為 48

$$4. |2x+5|=11$$

$$\Rightarrow 2x+5=\pm 11$$

$$\Rightarrow 2x=\pm 11-5=6 \text{ 或 } -16$$

$$\Rightarrow x=3 \text{ 或 } -8$$

$$5. |2x+1| \leq 5$$

$$\Rightarrow -5 \leq 2x+1 \leq 5 \xrightarrow{-1} -6 \leq 2x \leq 4$$

$$\xrightarrow{+2} -3 \leq x \leq 2$$

$$6. \text{ 令 } x=-1 \text{ 代入 } y=x^2: y=(-1)^2=1$$

則 P 點為 $(-1,1)$

$$\text{令 } x=2 \text{ 代入 } y=x^2: y=2^2=4$$

則 Q 點為 $(2,4)$

故 P 和 Q 的距離為

$$\sqrt{[2-(-1)]^2 + (4-1)^2} = \sqrt{18} = 3\sqrt{2}$$

$$7. \because \overline{AP} : \overline{PB} = 3 : 1$$

$$\therefore P = \frac{3B+1A}{3+1} = \frac{3(2,2)+(0,0)}{4}$$

$$= \frac{(6,6)+(0,0)}{4} = \frac{(6,6)}{4} = \left(\frac{3}{2}, \frac{3}{2}\right)$$

$$\text{故 } m = \frac{3}{2}, n = \frac{3}{2}, \text{ 因此 } m+n = \frac{3}{2} + \frac{3}{2} = 3$$

$$8. \text{ 令 } x=-1 \text{ 代入 } y=x^2+1: y=(-1)^2+1=2$$

則 P 點為 $(-1,2)$

$$\text{令 } x=3 \text{ 代入 } y=x^2+1: y=3^2+1=10$$

則 Q 點為 $(3,10)$

$\overline{PQ}$ 中點

$$= \frac{P+Q}{2} = \frac{(-1,2)+(3,10)}{2} = \frac{(2,12)}{2} = (1,6)$$

$$\text{故 } a=1, b=6, \text{ 因此 } a+b=1+6=7$$

$$9. \text{ 平行四邊形 } ABCD \text{ 中, } A+C=B+D$$

$$\text{則 } D=A+C-B=(5,2)+(-4,3)-(1,3)=(0,2)$$

故 D 點為 $(0,2)$

$$10. \text{ 重心} = \frac{A+B+C}{3} = \frac{(3,-5)+(-1,8)+(7,6)}{3}$$

$$= \frac{(9,9)}{3} = (3,3)$$

② 單元 1 坐標系與函數圖形

(主題 14~23：函數及其圖形、一元二次不等式)

答案

1. 37
2. 2
3. 三
4. -3
5. (1, -1)
6. $-\frac{4}{5}$
7. $x \geq -2$
8. 2
9. $x < -\frac{1}{7}$ 或 $x > 7$
10. 無實數解

1. $f(2) = 3 \times 2^3 + 5 = 29$

$f(0) = 8$

故 $f(2) + f(0) = 29 + 8 = 37$

2. $f(-2) = 2 \times (-2)^2 + a \times (-2) + 5 = 13 - 2a$

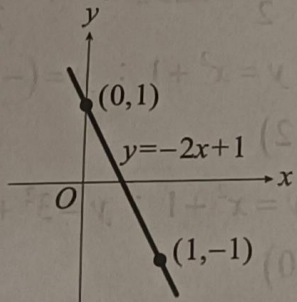
$f(1) = 2 \times 1^2 + a \times 1 + 5 = 7 + a$

$\therefore f(-2) = f(1)$

$\therefore 13 - 2a = 7 + a \Rightarrow -3a = -6 \Rightarrow a = 2$

3. 令 $y = -2x + 1$, $\begin{array}{c|c|c} x & 0 & 1 \\ \hline y & 1 & -1 \end{array}$

其圖形如圖



故此函數的圖形不會經過第三象限

4. $\therefore$ 拋物線通過點 $(-1, 0)$

$\therefore 0 = (-1)^2 + a \times (-1) + b$

$\Rightarrow -a + b = -1 \dots \dots \textcircled{1}$

$\therefore$ 拋物線通過點 $(2, 0)$

$\therefore 0 = 2^2 + a \times 2 + b \Rightarrow 2a + b = -4 \dots \dots \textcircled{2}$

$\textcircled{2} - \textcircled{1} : 3a = -3 \Rightarrow a = -1$

$a = -1$ 代回 $\textcircled{1} : -(-1) + b = -1 \Rightarrow b = -2$

故 $a + b = -1 + (-2) = -3$

5. $y = 2x^2 - 4x + 1$

$= 2(x^2 - 2x + 1) + 1 - 2 = 2(x - 1)^2 - 1$

故此二次函數的頂點為 $(1, -1)$

6. 令 $a = 5, b = 6, c = 1$

最小值 $= \frac{4ac - b^2}{4a} = \frac{4 \times 5 \times 1 - 6^2}{4 \times 5}$
 $= \frac{20 - 36}{20} = \frac{-16}{20} = -\frac{4}{5}$

7. 原式: $-2x - 14 \leq 4x - 2$

$\Rightarrow -2x - 4x \leq -2 + 14$

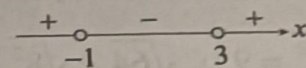
$\Rightarrow -6x \leq 12 \xrightarrow{+(-6)} \Rightarrow x \geq -2$

8. 因式分解 $x^2 - 2x - 3 = (x - 3)(x + 1)$

令 $(x - 3)(x + 1) = 0 \Rightarrow x = 3$ 或 -1

則 $x^2 - 2x - 3 < 0$ 的解為 $-1 < x < 3$

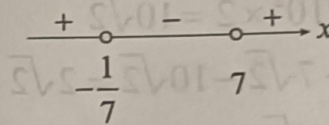
故 $a = -1, b = 3$, 因此 $a + b = -1 + 3 = 2$



9. 因式分解 $7x^2 - 48x - 7 = (7x + 1)(x - 7)$

令 $(7x + 1)(x - 7) = 0 \Rightarrow x = -\frac{1}{7}$ 或 7

則 $7x^2 - 48x - 7 > 0$ 的解為 $x < -\frac{1}{7}$ 或 $x > 7$



10. $\therefore x^2 - 6x + 9 = x^2 - 2 \times 3 \times x + 3^2 = (x - 3)^2$

$\Rightarrow$ 恆正或零

$\therefore x^2 - 6x + 9 < 0 \Rightarrow$ 無實數解